

Sumedh Kothari

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EDUCATION

Georgia Institute of Technology

B.S. in Chemical and Biomolecular Engineering

Minor in Computer Science - Intelligence

Minor in Quantum Sciences and Technology

Expected May 2028

GPA: 3.91

- **Coursework:** Computer Organization and Programming; Data Structures and Algorithms; Differential Equations
- **Honors:** PACESHIP Scholar, TechHack Valley 2026 Hackathon Winner, Symposium of Rising Scholars Presenter, Eagle Scout

EXPERIENCE

Witness.ai

Machine Learning Intern

Mountain View, CA

May - September 2026

- Designed and implemented a novel adversarial machine learning framework combining GANs, SFT, GRPO, and RLVR to autonomously generate harmful prompts for discovering jailbreak vulnerabilities in modern LLMs (Claude Haiku, Qwen) and reaching 60% ASR, optimizing attack generation policies through iterative reward-based learning with ongoing work toward arXiv publication
- Engineered an end-to-end agentic research pipeline in Python using PyTorch, vLLM, and HuggingFace that autonomously processed 1,000+ arXiv papers, synthesized emerging LLM red-teaming methodologies, and generated novel adversarial attack strategies
- Built a scalable adversarial evaluation framework that generated 3,000+ adversarial prompts, benchmarked 8 state-of-the-art open-source and proprietary LLMs, fine-tuned 3 LLM safety classifiers on synthesized data, and produced automated customer-facing safety reports

ChemE Cube | Georgia Tech

Software & Mechanical Lead

Atlanta, GA

August 2025 - Present

- Developed a Gaussian-process Bayesian optimization framework for CO₂ capture experiments, learning a surrogate model of capture performance from experimental data and selecting high-value operating conditions to accelerate reactor optimization
- Leading the mechanical development of a 1 ft³ direct air capture reactor by managing a 5-member subteam and overseeing end-to-end CAD, prototyping, and systems integration, delivering a fully fabricated and test-ready prototype
- Designing airflow distribution and sorbent containment subsystems using parametric CAD, 3D printing, and CNC machining, achieving 75% reduction of ambient CO₂ in bench-scale validation for AIChE competition (Nov 2026)

RESEARCH

Ramprasad Laboratory | Georgia Tech

Machine Learning Researcher

Atlanta, GA

July 2026 - Present

- *MLFFs for Polymers* — Two-Body Pairing Message Parsing Graph Neural Network for Accurate, Efficient Polymer DFT Surrogate

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Machine Learning Researcher

Mountain View, CA

May - September 2026

- *Adversarial Self-Play for Automated LLM Jailbreak Optimization* — Joint GRPO, SFT, RLVR, Generative Adversarial Training
- *LiLAC: A Living Literature Adversarial Corpus for Evaluating and Improving Jailbreak Guards* — Introduces an automated framework for continuously converting emerging jailbreak research into adversarial data for training and evaluating LLM safety classifiers.

PROJECTS

POLYT5-RLVR: Polymer Encoder-Decoder Model

Machine Learning Researcher

Atlanta, GA

August 2026 - Present

- Reproduced and post-trained PolyT5, a 7.5M parameter T5 pretrained on 92.3M polymers, and cut invalid polymer generations by 79% via GRPO and RL against chemistry-verifiable rewards
- Cut Tg prediction error 5.2% (28.60 → 27.11 K) and structurally eliminated catastrophic decode failures by replacing text decoding with a linear regression head, validated by a two-seed unanimity criterion cleared on 10/10 splits across a 70-run five-factor ablation
- Shipped distribution-free split-conformal prediction intervals (± 54.35 K at 90%), converting point predictions into calibrated ranges

Machine Learning-Assisted Particle Swarm Optimization

Machine Learning Researcher

Atlanta, GA

March - May 2026

- Awarded a \$1,000 PACESHIP Scholar research grant to investigate machine learning methods for black-box optimization, conducting research on surrogate-assisted Particle Swarm Optimization to improve optimization efficiency in high-dimensional search spaces
- Designed and implemented a hybrid optimization framework integrating Gaussian Process surrogate modeling, machine learning-guided particle repositioning using LightGBM model, and adaptive exploration-exploitation strategies to identify stagnating particles and dynamically steer search trajectories toward promising solution regions
- Developed an autoresearch framework inspired by iterative AI research agents that autonomously executed optimization experiments, adjusted hyperparameters, analyzed performance trends, and refined search strategies through closed-loop experimentation; built a scalable PyTorch pipeline achieving an average 28.8% improvement over baseline PSO across benchmark functions, including 97.8% on Sphere, 99.9% on Rastrigin, 90.4% on Ackley, and 58.8% on Rosenbrock landscapes

SportSplat: NFL 4D Gaussian Splatting

Lead Developer

Atlanta, GA

February 2026 - Present

- Developing a 4D Gaussian Splatting-based volumetric reconstruction pipeline to transform multi-view third-person sports footage into temporally coherent, player-specific first-person renderings, optimizing dynamic representation with spatiotemporal Gaussian primitives
- Training ML model to apply 3D Gaussians onto SMPL-X human frames to recreate NFL video footage from any player perspective
- Deploying a differentiable rendering pipeline with camera pose optimization (PyTorch/CUDA) to reconstruct dynamic 3D player trajectories, and optimize Gaussian parameters for real-time, personalized first-person sports video synthesis

SKILLS

- **Technical:** Python, PyTorch, NumPy, Pandas, Scikit-learn, CUDA, Linux, DFT, MATLAB, CAD (SolidWorks, Fusion)
- **Communication:** Technical Writing, Multilingual (English, Hindi, Spanish)